THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

IRINGA, LINDI AND MOROGORO FORM FOUR REGIONAL MOCK EXAMINATIONS, 2024

CODE: 032/2A

CHEMISTRY 2A
ACTUAL PRACTICAL

TIME: 3 HOURS **May, 2024**

INSTRUCTIONS

1. This paper consists of two (2) questions. Answer all the questions.
2. Each question carries twenty-five (25) marks.
3. Cellular phones and any unauthorized materials are not allowed in the examination room.
4. Write your **examination number** on every page of your answer sheet(s).
5. Qualitative analysis sheet may be used
6. You may use the following constants, where necessary
   Atomic masses: H=1, O=16, Na=23, S=32, Cl=35.5, C=12
   1litre=1dm^3=1000cm^3

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1. You are provided with the following:
   H: A 3.65g of pure hydrochloric acid in a litre of solution
   F: A 3.50g of impure sodium hydroxide in 500cm^3 solution
   Methyl orange indicator

   Procedure

   Titrate the acid (in a burette) against the base (in a conical flask) using two drops of your indicator to obtain three titre values.

   Questions

   (a) (i).....cm^3 of **H** required .....cm^3 of **F** for complete neutralization

   (ii) The colour change during titration was from.....to.....

   (iii) Is it possible to use Phenolphthalein indicator in place of methyl orange in this particular experiment?

   (b) Write the balanced reaction equation for this experiment.

   (c) Calculate the percentage impurity of sodium hydroxide

   (d) What is the name of solution placed in the

   (i) Burette

   (ii) Conical flask

   (e) Is the end-point or equivalence point reached first during titration?

2. Carefully carry out the experiments described below and record all your observations, inferences and hence identify the cation and anion present in sample salt **G**

S/NO.	Experiment	Observation	Inference
(a)	Observe sample G		
(b)	Take a sample G in a glass rod /back of the test tube and dip into Concentrated hydrochloric acid and heat in a non-luminous flame.		
(c)	Heat a little amount of sample G into a dry test tube, heat gently and then strongly.		
(d)	Put a little amount of sample G into a test – tube and divide the solution into four portions (four test tubes) (i) Into the first test tube add small amount of ammonia solution followed by ammonium oxalate solution. (ii) Into the first test tube add Sodium hydroxide solution dropwise until in excess. (iii) Into the third test tube add small amount of MnO_2 followed by small amount of concentrated Sulphuric acid and warm the mixture (iv) Into the fourth test-tube add small drops of silver nitrate until in excess.		

QUESTIONS

(a) The cation present in sample G is_____and the anion is_____

(b) Name sample G_____

(c) Write the balanced chemical equations for the reaction taking place in experiment d (i)above

(d) What are the uses (three uses) of anion present in sample G

(e) What is the type of the bond formed between anion and cation of sample G.?